



Accelerated modelling of moisture diffusion controlled drying using coupled physics informed neural network

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ABSTRACT

A coupled physics informed neural network (CPINN) was used to simulate liquid diffusion controlled drying, an energy intensive process in the food industry. The architecture of the CPINN was designed to permit the prediction of thermo-physical properties and key source and sink terms at the solution boundaries which cause the solution to be highly coupled. The CPINN structure improves upon limitations of using PINNs in low-temperature food drying simulations, most notably allowing multiple and highly coupled variables to be simulated in addition to ensuring dynamic thermo-physical properties updates. The CPINN successfully solved a system 1-D partial differential equations (PDEs), capturing phenomena such as transient moisture diffusion and heat conduction, evaporative and convective heat transfer at the drying surface and moisture loss to the drying air. A benchmark simulation was used to compare the CPINN predicted product temperature, $\hat{T}_p$, and predicted moisture content, $\hat{X}_p$, against a numeric solution. The mean absolute error for the respective comparisons was 0.12 °C and 0.0035 kg_m kg_s⁻¹. Training the CPINN for the first time was the rate limiting step, requiring the greatest time to solve when compared to the numeric solution, with solution times of $t_{cpinn} = 321$ min and $t_{rk} = 82.7$ min, respectively, or a time reduction fraction of $t_r = 3.9$, due to generalised initialisation of the CPINN parameters. By utilising a staged transfer learning approach, t_r was reduced to a range of 0.28–0.027 whilst maintaining solution accuracy, representing a 3 to 37 times faster solution. By saving a library of CPINN models, solutions at key drying conditions of interest can be rapidly evaluated at run time, meaning the saved CPINN effectively acted as a method to compress solutions of PDEs. The techniques used here show how CPINNs can be applied to coupled and multi-scale PDEs using a physics-based approach to problems in the food processing and other sectors.

1. Introduction

Solution to common industrial problems often require evaluation of ordinary (ODEs) and partial differential equations (PDEs), for example when analysing systems involving fluid flow, heat transfer or chemical reaction. The solution methods employed are typically numeric or iterative in nature (Versteeg and Malalasekera, 1995). They have significant computational burden and can take a significant amount of time to solve depending on the system complexity, stability and dimensionality (Dobbelaere et al., 2021). Alternatively, the solution may be derived from experimentation, which is often too costly or does not truly represent the system in question. For example experimentation may not be able to capture the scale of the real system. Similarly,

analytical solutions may be unknown for complex systems and must be simplified with numerous assumptions to be solvable (Shirvany et al., 2009).

The simulation of large scale processes in a reasonable time frame whilst maintaining solution accuracy remains a significant challenge. Results from these simulations are directly used in the optimisation of industrial processes, leading to improvements of various key performance indicators including economic, quality, environmental and safety factors (Gil Chaves et al., 2016). One such large scale and energy intensive process is drying food products, which requires the removal of moisture via heat transfer to facilitate the mass transfer of water out of a drying product (Raisul Islam et al., 2003). Optimising design

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through simulation helps to minimise energy consumption and maximise product quality and quantity. However, due to limited compute power, the non-linear and coupled nature of heat and mass transfer occurring during drying and the large number of variables to optimise, finding a global optimum using numerical simulations is challenging.

More recently, physics informed neural networks (PINNs) (Raissi et al., 2019; Lagaris et al., 1998) have been used as an alternative method to solve linear and nonlinear PDEs. PINNs are commonly constructed using deep feed-forward neural networks and the parameters that provide the mapping from inputs, e.g. time and space, to outputs, e.g. velocity, pressure or temperature, are trained on the fundamental physical laws described by the PDEs. The loss functions are summations of residual errors, typically calculated at sampled collocation points within the solution domain using the PDEs, and at sampled boundary points for the initial and boundary conditions or expressions. By minimising the summation of domain and boundary loss functions through finding optimal network parameters, a correct solution is obtained.

This methodology has certain advantages over traditional numeric methods, especially compared to explicit methods. The solution using PINNs is easily parallelisable over multiple input variables, for explicit methods, the solution is solved sequentially at each solution point. In addition, unlike numeric methods, no discretisation or meshing is required; the PINNs are trained on numerous sampled points which can be generated as required. Finally, known experimental or archived simulation data can be included into the training of the PINN (Raissi et al., 2019). PINNs have been successfully applied to various problems including, heat transfer problems (Zobeiry and Humfeld, 2021), the transient viscous burgers equation for gas dynamics problems (Meng et al., 2020) and various transient and two dimensional diffusion equations with different forcing functions (Meng et al., 2020; Fang et al., 2023).

The use of deep learning for modelling food drying processes has been previously studied without a physics based approach (Khan et al., 2022). For example Islam et al. (2003) used the numeric solution of a liquid diffusion model to generate 256 drying curves and trained a feed-forward network to rapidly predict drying curves using the Page equation. Although this provided rapid prediction, training of the network relied on numerical solution data, representing a rate limiting step. Other studies have relied on experimental data to solve the problem of predicting temperature and moisture curves for drying of cassava and mango (Hernandez-Perez et al., 2004) and point estimates of drying time of shelled corn (Momenzadeh et al., 2011), for example. Both experimentation and numeric solution have a significant time burden, especially when trying to optimise numerous key drying conditions and generate output training data.

Therefore, in this study the focus will be on simulating the drying problem with a coupled PINNs (CPINNs), removing the requirement of data collection through numerical studies or experimentation. A CPINN provides a neural network for each output variable, which are then coupled through the calculation of thermo-physical properties, derivatives and loss function calculations which constrains the solution to the underlying physics of the problem. CPINNs have been used previously to model the thermochemical curing of composite materials (Amini Niaki et al., 2021), where two neural networks were used to predict composite temperature and degree of cure, where each variable was interdependent. However, unlike the CPINN in this study, the thermo-physical properties were not predicted by the network. Additionally, a coupled system has been used to predicting polymer microphase-separated structures, utilising networks in a staged approach to provide initial guesses for key variables in system of diffusion equations (Wei et al., 2018). In the domain of low-temperature food drying, PINNs have been used successfully to predict the moisture mass transfer within plant cells, neglecting the change of temperature within the drying product and any changes to the thermo-physical properties (Batuwatta-Gamage et al., 2022). In this study, the limitations of using PINNs in low-temperature food drying simulations are addressed, by not only

solving for multiple and highly coupled variables using a coupled architecture but also by ensuring the thermo-physical properties are also successfully predicted by the CPINN, ensuring a fully dynamic simulation.

This work contributions to solving coupled PDEs for food drying processes using physics informed deep learning. Specifically, a system of CPINNs is used to simulate a typical industrial drying process based on convective drying of a food product using low temperature air in a performant manner. This is a common method of moisture removal for applications such as food preservation (Sinha and Bhargav, 2022). The CPINNs provide predictions of product temperature and moisture content as a function of time and drying product thickness. The CPINNs solution is compared to a numerical method using simulation error and solution time as performance benchmarks. The CPINN architecture is constructed to additionally provide predictions of thermo-physical properties. Through automatic differentiation, relevant gradients and fluxes at the drying surface of the food product are also obtained. Therefore, due to the structure and training process, the CPINNs permitted the variability of the food product properties and drying conditions to be captured. Finally, transfer learning is used to provide a reduction in solution time when compared to explicit numeric methods for simulation of different key drying conditions, in this case, initial moisture content and drying air temperature. A library of pre-trained CPINN at each key drying condition is saved for future evaluation, acting as solution generators which can be evaluated at run time.

2. Drying model

The drying model employed in this work was adapted from the system of PDEs proposed by Raisul Islam et al. (2003) to describe the change in product temperature, T_p , and moisture density, ρ_m , as a function of time, t , and product depth, z . The two-dimensional case, considering drying additionally over the product width was deemed computationally inefficient due to the high aspect ratios common in food drying processes. The model describes a drying product, in this case a potato was used as the model material, which is initially saturated with a given amount of moisture as specified by an initial moisture content, $X_{m,t=0}$. The product is dried under a constant flow of low temperature air, where moisture is lost at the drying surface and heat is transferred by convection from the air to the product and by an evaporative flux of water at the surface as moisture leaves the product as vapour. A depiction of this process is shown in Fig. 1. The equations, conditions and assumptions are applicable for the numeric method and coupled PINN solutions (Raisul Islam et al., 2003; Islam et al., 2003):

1. Drying is considered as a one-dimensional and transient problem, where heat and moisture vary only along the drying product depth, i.e. $z = 0$ mm to L .
2. The base plate is impermeable to moisture and adiabatic.
3. The drying product is homogeneous and without voids.
4. The drying product does not shrink during drying.
5. The physical properties are calculated as empirical correlations, shown in Table 1, and were dependent on solution moisture and temperature.
6. Evaporation of water only occurs at the drying surface and not within the drying product.

2.1. Domain expressions

The transient energy balance, describing T_p change over t and z via conduction only was defined by Eq. (1). The product thermal conductivity, k_p , and density, ρ_p , were dependent on the solution variables and the product specific heat capacity, C_p , was held constant as specified in Table 1. The transient mass balance, describing ρ_m change over t and z via diffusion only was defined by Eq. (2). The moisture diffusivity, D_w , was dependent on the solution variables and is specified in Table 1.

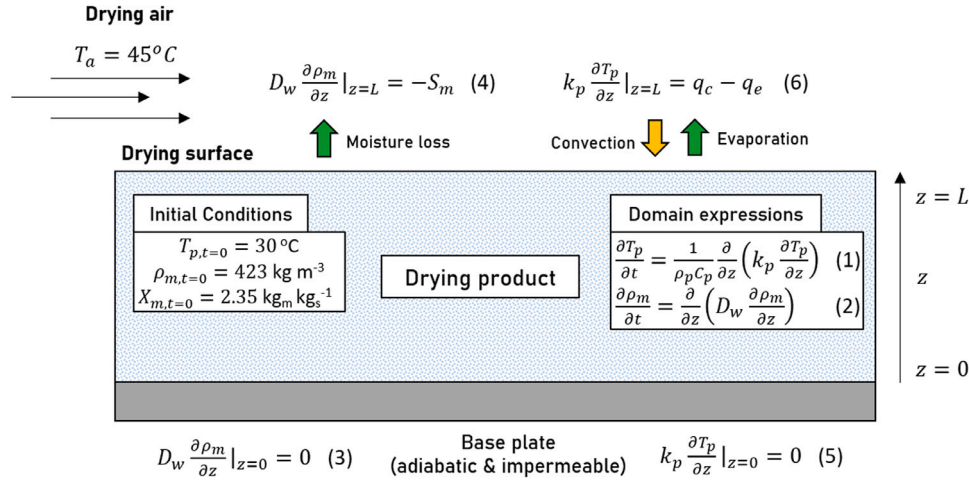


Fig. 1. Simulation geometry, including expressions that describe the change in T_p and ρ_m as a function of t and z . Boundary conditions are specified for the base plate and at the drying surface. At the drying surface, moisture is lost based on a surface mass flux and heat is transferred by forced convection due to low temperature drying air and due to evaporation of moisture.

Table 1

Thermal and physical properties (Raisul Islam et al., 2003; Keey, 1978). Symbols defined in the nomenclature.

Property	Range	Correlation
Thermal properties		
h_{fg} [kJ kg ⁻¹]		$-2.394 T_p + 2502.1$
h_v [kJ kg ⁻¹]		$h_{fg} + \Delta h_w$
Δh_w [kJ kg ⁻¹]	$X_m \leq 0.02$	1420.95
	$0.02 \leq X_m \leq 0.2$	$-8.2068 \cdot 10^6 X_m^4 + 4.0003 \cdot 10^6 X_m^3 - 6.1614 \cdot 10^5 X_m^2 + 2.3682 \cdot 10^4 X_m + 1.1631 \cdot 10^3$
	$X_m > 0.2$	$-22.583 X_m + 130.18$
k_p [W m ⁻¹ K ⁻¹]		$\frac{0.049}{1+X_m} \cdot \exp \left[\frac{-47}{8.3143 \cdot 10^{-3}} \cdot \left(\frac{1}{T_p+273.15} - \frac{1}{333.15} \right) \right] + \frac{0.611 X_m}{1+X_m}$
k_a [W m ⁻¹ K ⁻¹]		$6.818 \cdot 10^{-10} T_a^3 - 1.474 \cdot 10^{-7} T_a^2 + 8.029 \cdot 10^{-5} T_a + 0.02408$
Nu [-]		$0.664 Re^{0.5} Pr^{0.33}$
Pr [-]		$-2.273 \cdot 10^{-8} T_a^3 + 4.199 \cdot 10^{-6} T_a^2 - 3.534 \cdot 10^{-4} T_a + 0.719$
Physical properties		
A_w [-]	$X_m < 0.22$	$1232.3 \cdot X_m^4 - 646.14 \cdot X_m^3 + 98.541 \cdot X_m^2 - 0.3406 \cdot X_m - 0.0008$
	$0.22 \leq X_m \leq 0.6$	$-0.5129 \cdot X_m^2 + 0.6398 X_m + 0.5995$
	$X_m > 0.6$	$0.0013 \cdot X_m^3 - 0.0197 \cdot X_m^2 + 0.1164 X_m + 0.7448$
D_w [m ² s ⁻¹]		$1.29 \cdot 10^{-6} \exp \left[\frac{-0.0725}{X_m} \right] \exp \left[\frac{-2044}{T_p+273.15} \right]$
P_{sat} [Pa]		$100 \cdot \exp \left[27.0214 - \frac{6887}{T+273.15} - 5.31 \cdot \ln \left(\frac{T+273.15}{273.15} \right) \right]$
Sh [-]		$0.664 Re^{0.5} Sc^{0.33}$
ρ_a [kg m ⁻³]		$-3.510 \cdot 10^{-8} T_a^3 + 1.584 \cdot 10^{-5} T_a^2 - 4.670 \cdot 10^{-3} T_a + 1.2921$
ρ_p [kg m ⁻³]		$-1 \cdot 10^{-3} X_m^4 - 6.4 X_m^3 + 55 X_m^2 - 154 X_m + 1253$
μ_a [Pa s]		$1.768 \cdot 10^{-13} T_a^3 - 5.541 \cdot 10^{-11} T_a^2 + 4.983 \cdot 10^{-8} T_a + 17.1964 \cdot 10^{-6}$

2.2. Initial, boundary and drying conditions

Conditions for the drying air and the product properties, including the initial conditions used in the numeric and CPINN models, are summarised in Table 2. Specifically, the $X_{m,t=0}$ and initial product temperature, $T_{p,t=0}$, were used to initialise the models.

Mass and heat transfer boundary conditions were specified at the base of the drying product ($z = 0$ mm), as defined by Eq. (3) and Eq. (5), respectively, to represent the impermeable and adiabatic nature of the base plate. At the drying surface ($z = L$) moisture loss was described by a diffusive mass flux (Eq. (4)) and moisture sink term, S_m , whilst heat transfer was described by conductive heat transfer (Eq. (6)), with terms to represent heat transfer by convection, q_c , and due to evaporation of moisture, q_e . S_m represents the difference between the amount moisture at the surface of the drying product and the amount of moisture in the

drying air as shown by Eq. (7).

$$S_m = \frac{h_m}{M_w R} \left(\frac{A_{w,z=L} P_{sat,z=L}}{T_{p,z=L}} - \frac{\psi P_{sat,a}}{T_a} \right) \quad (7)$$

The amount of moisture at the surface of the drying product was evaluated using the saturation vapour pressure at the drying surface, $P_{sat,z=L}$, water activity, A_w , and product temperature at the drying surface, $T_{p,z=L}$. The amount of moisture in the drying air was evaluated using the saturation vapour pressure of air, $P_{sat,a}$, relative humidity, ψ , and air temperature, T_a . The mass transfer coefficient of moisture at the drying surface, h_m , was determined from the Sherwood number, Sh , as defined in Table 1 and M_w and R were simply the molecular weight of water and the ideal gas constant.

The heat transfer at the drying surface was controlled by convective heat transfer from the drying air and evaporation of moisture which

Table 2

Drying conditions, product properties and simulation specifications (Raisul Islam et al., 2003). Bold values indicate simulation conditions for the benchmark solution.

Drying conditions	
Air temperature, T_a [°C]	40.0, 42.5, 45.0
Air velocity, u_a [m s ⁻¹]	1.5
Relative humidity, ψ [%]	10
Product properties and initial conditions	
Product specific heat capacity, C_p [kJ kg ⁻¹ K ⁻¹]	1660
Dry product density, ρ_s [kg, m ⁻³]	180
Initial moisture content, $X_{m,t=0}$ [kg _m kg _s ⁻¹]	2.00, 2.15, 2.35
Initial moisture density, $\rho_{m,t=0}$ [kg _m m ⁻³]	360, 387, 432
Initial product temperature, $T_{p,t=0}$ [°C]	30
Simulations conditions and bounds	
Final simulation time, t_f [min]	400
Drying product thickness, L [mm]	5

were defined by Eqs. (8) and (9), respectively. The air heat transfer coefficient, h_a , was determined using the Nusselt number, Nu , which was dependent on the air properties, air velocity, u_a , and T_a . The latent heat of vaporisation, h_{fg} , and heat of wetting, Δh_w , summed to the total enthalpy of vaporisation. All these parameters were based on correlations summarised in Table 1.

$$q_c = h_a(T_a - T_{p,z=L}) \quad (8)$$

$$q_e = S_m h_{v,z=L} \quad (9)$$

2.3. Numerical solution

The benchmark numeric solution was obtained using a fourth order Runge Kutta method. The PDEs were discretised using central differencing for the spatial derivative as outlined by Eq. (10) (Yeung and Lam, 1996). The spatial derivatives were written in a vectorised format using the NumPy scientific computing library to ensure performant computation.

To assess accuracy of this solution and ensure correct error comparison against the system of coupled PINNs a mesh and time step study was completed. The mesh study was achieved by first specifying a sufficiently small time step and then decreasing the discretised domain size until the solution no longer changed within an absolute mean error tolerance. The time step was chosen as 0.0004 s and the domain size for a consistent solution was 0.025 mm. Once the mesh study was completed, the time-step was increased until the solution began to deviate from the accurate solution, the largest time-step for an accurate solution was chosen, which was 0.0010 s, providing an optimised mesh and time-step for performance comparisons against the coupled PINN model. Results of the mesh study can be found in the supporting information.

$$\frac{\partial}{\partial z} \left(k_p \frac{\partial T}{\partial z} \right) = \frac{k_{p,z+1} (T_{p,z+1} - T_{p,z-1}) + 4k_{p,z} (T_{p,z+1} - 2T_{p,z} + T_{p,z-1}) + k_{p,z-1} (T_{p,z-1} - T_{p,z+1})}{4(dz)^2} \quad (10)$$

2.4. Coupled physics informed neural networks for drying processes

To solve the system of equations a CPINN was constructed using the network architecture as depicted in Fig. 2. As found in previous studies (Amini Niaki et al., 2021), single PINNs trained for each solution variable provides improved performance when compared to a single network approximating both variables simultaneously. Therefore, two deep feed-forward neural networks were used in this work, one to solve for the approximate product temperature, $\hat{T}_p$, defined as $\mathcal{N}_T$, and one to solve for the approximate product moisture density, $\hat{\rho}_m$, defined as $\mathcal{N}_\rho$, represented by Eqs. (11) and (12), respectively. Input

variables to the network were collections of sampled t and z points, bounded by the boundary and initial limits; sampling is described in Section 2.5. Each network provided a non-linear mapping between the input variables and the output solutions. A correct solution was found by CPINN training, which involved learning the optimal network parameters, θ^* , through physics informed loss functions to provide a correct solution to the drying problem at the domain and boundary points (Ian Goodfellow et al., 2016). The optimisation procedure can be found at the end of this section as is formally defined by Eqs. (25) and (26).

$$T_p \approx \hat{T}_p = \mathcal{N}_T(t, z; \theta_T), \quad t \in [0, 400 \text{ min}], \quad z \in [0, 5 \text{ mm}] \quad (11)$$

$$\rho_m \approx \hat{\rho}_m = \mathcal{N}_\rho(t, z; \theta_\rho), \quad t \in [0, 400 \text{ min}], \quad z \in [0, 5 \text{ mm}] \quad (12)$$

The deep feed-forward neural networks are composed of repeated functions (Eq. (13)), starting from input nodes, which for both networks were $[t, z]$ and the output node was either $\hat{T}_p$ or $\hat{\rho}_m$ for the drying model.

$$\mathcal{N}(t, z; \theta) = f^I \dots (f^{i+1}(f^i(f^0))), \quad i = 1, \dots, I \quad (13)$$

The resultant values at the middle nodes, known as hidden nodes of the network, are calculated by a linear operation, multiplying the current node inputs by weight matrices, $\mathbf{W}^i$, and addition of bias vectors, $\mathbf{b}^i$, together forming the trainable network parameters, θ , (Amini Niaki et al., 2021; Russel and Norvig, 2022; Ian Goodfellow et al., 2016). A non-linear activation function, σ , is then applied to the linear operation to provide a non-linear mapping of inputs to outputs; the combination of these functions is summarised by Eq. (14). As it is necessary to compute first and second order derivatives of the feed-forward networks for the physics informed loss functions, the hyperbolic tangent activation function was chosen as the second order derivative does not evaluate to zero (Zobeiry and Humfeld, 2021).

$$f^i = \sigma(\mathbf{W}^i \cdot f^{i-1} + \mathbf{b}^i), \quad i = 1, \dots, I \quad (14)$$

For each CPINN, loss functions are defined which express the deviation of the current approximation from the physical laws applied by the drying model in the domain, at the boundaries and at the initial condition. Two loss function were imposed, one for the heat balance and one for the moisture balance. The heat balance loss function, L_T , (Eq. (19)) was the summation of the heat domain, $L_{T,d}$, (Eq. (15)) inlet condition, $L_{T,z=0}$, (Eq. (16)) outlet condition, $L_{T,z=L}$, (Eq. (17)) and initial condition, $L_{T,t=0}$, (Eq. (18)) loss functions. The heat domain loss function was the residual of the transient heat balance equation (Eq. (1)), expressed as a mean squared error (MSE). Similarly, the boundary conditions were imposed by the respective residual as described by Eqs. (5) and (6). $L_{T,z=L}$ included the contributions of heat loss due to convection and evaporation. For both the inlet and outlet conditions, thermal properties and derivatives were evaluated at the sampled boundary points at $z = 0.0$ mm and $z = L$, respectively. The approximate first and second order differentials of T_p were found by automatic differentiation (Baydin et al., 2018). The initial condition was simply a MSE error comparing the known $T_{p,t=0}$ to the network prediction, $\hat{T}_{p,t=0}$.

$$L_{T,d} = \frac{1}{N_d} \sum_i \left[\frac{\partial \hat{T}_p}{\partial t} - \frac{1}{C_p \hat{\rho}_p} \frac{\partial}{\partial z} \left(\hat{k}_p \frac{\partial \hat{T}_p}{\partial z} \right) \right]^2 \quad (15)$$

$$L_{T,z=0} = \frac{\lambda_{T,z=0}}{N_{z=0}} \sum_i \left[-\hat{k}_{p,z=0} \frac{\partial \hat{T}_p}{\partial z} \right]_{z=0}^2 \quad (16)$$

$$L_{T,z=L} = \frac{\lambda_{T,z=L}}{N_{z=L}} \sum_i \left[h_p(T_a - \hat{T}_{p,z=L}) - \hat{S}_m \hat{h}_{v,z=L} - \hat{k}_{p,z=L} \frac{\partial \hat{T}_p}{\partial z} \right]_{z=L}^2 \quad (17)$$

$$L_{T,t=0} = \frac{\lambda_{T,t=0}}{N_{t=0}} \sum_i [T_{p,t=0} - \hat{T}_{p,t=0}]^2 \quad (18)$$

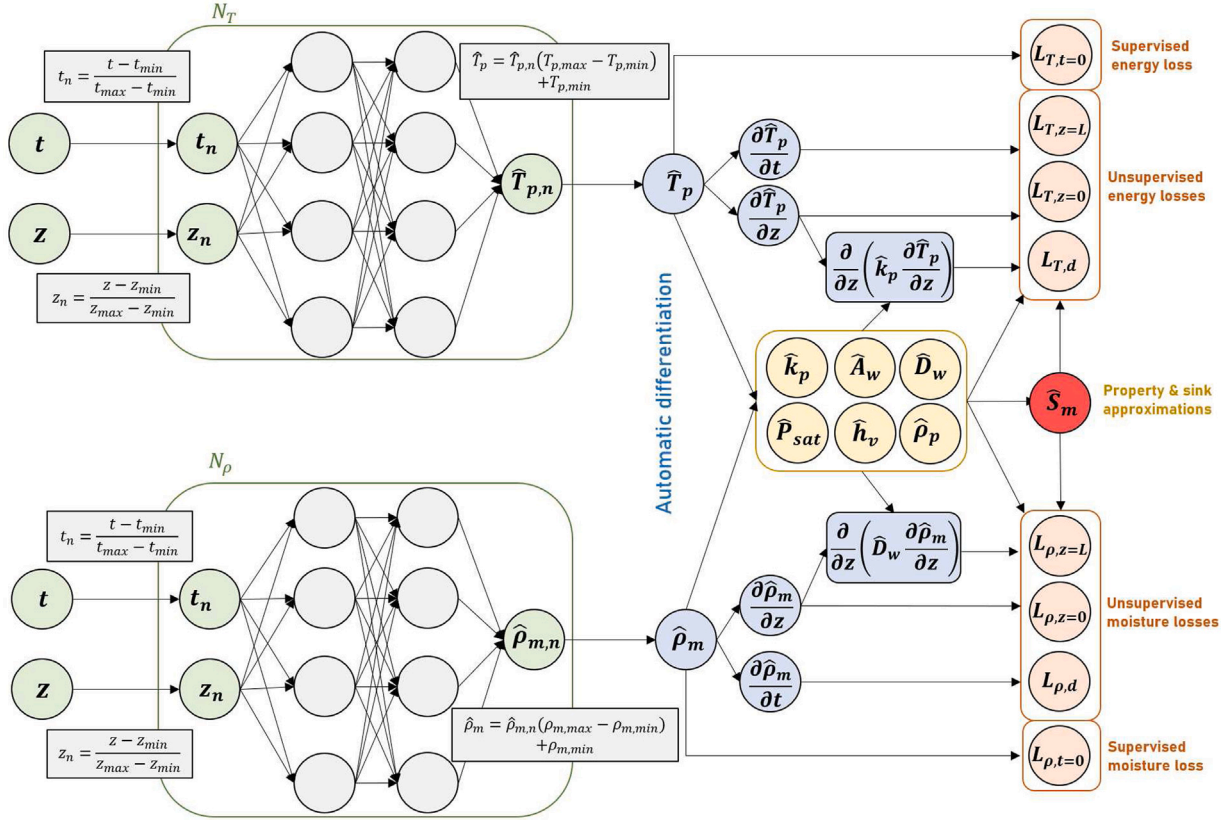


Fig. 2. Architecture of the coupled physics informed neural network (CPINN). Two networks are present, one which provides a prediction for $\hat{T}_p$ and associated derivatives and thermo-physical properties for the heat balance loss functions and the other provides a prediction for $\hat{\rho}_m$ and associated derivatives and thermo-physical properties for the moisture balance loss functions. The structure shows the coupling of the two problems through the thermo-physical property estimations and $\hat{S}_m$.

$$L_T = L_{T,d} + L_{T,z=0} + L_{T,z=L} + L_{T,t=0} \quad (19)$$

The moisture loss function, L_ρ , (Eq. (24)) was the summation of the moisture domain, $L_{\rho,d}$, (Eq. (20)) inlet condition, $L_{\rho,z=0}$, (Eq. (21)) outlet condition, $L_{\rho,z=L}$, (Eq. (22)) and initial condition, $L_{\rho,t=0}$, (Eq. (23)) loss functions. The moisture domain loss function was the residual of the transient heat balance equation (Eq. (2)), expressed as a mean squared error (MSE). Similarly, the boundary conditions were imposed by the respective residual as defined earlier by Eqs. (3) and (4). $L_{\rho,z=L}$ included the contributions of moisture loss due to evaporation at the drying surface. The initial condition was simply a MSE error comparing the known $X_{m,t=0}$ to the network prediction, $\hat{X}_{m,t=0}$.

$$L_{\rho,d} = \frac{1}{N_d} \sum_i \left[\frac{\partial \hat{\rho}_m}{\partial t} - \frac{\partial}{\partial z} \left(\hat{D}_w \frac{\partial \hat{\rho}_m}{\partial z} \right) \right]^2 \quad (20)$$

$$L_{\rho,z=0} = \frac{\lambda_{\rho,z=0}}{N_{z=0}} \sum_i \left[-\hat{D}_w \frac{\partial \hat{\rho}_m}{\partial z} \Big|_{z=0} \right]^2 \quad (21)$$

$$L_{\rho,z=L} = \frac{\lambda_{\rho,z=L}}{N_{z=L}} \sum_i \left[-\hat{S}_m - \hat{D}_w \frac{\partial \hat{\rho}_m}{\partial z} \Big|_{z=L} \right]^2 \quad (22)$$

$$L_{\rho,t=0} = \frac{\lambda_{\rho,t=0}}{N_{t=0}} \sum_i \left[\rho_{m,t=0} - \hat{\rho}_{m,t=0} \right]^2 \quad (23)$$

$$L_\rho = L_{\rho,d} + L_{\rho,z=0} + L_{\rho,z=L} + L_{\rho,t=0} \quad (24)$$

Due to order of magnitude variations in the loss functions, arising from the fact that each loss function was specified in a different set of units, for example the domain heat loss function had units of $K s^{-1}$ and the boundary domain loss function had units of $W m^{-2}$, regularisation terms, λ , were included for each of the loss functions. The regularisation terms were added to more strongly impose the boundary conditions to

determine the unique solutions of the system of equations (Sinha and Bhargava, 2022).

Once the loss functions were defined, the optimisation problem can be written as shown by Eqs. (25) and (26), where the loss functions are minimised by finding optimal network parameters for each network using the sampled input parameters, t and z , and predicted values from each network, $\hat{T}_p$ and $\hat{\rho}_m$; a process known as network training.

$$\theta_T^* = \min_{\theta_T} L_T(t, z, \hat{T}_p, \hat{\rho}_m; \theta_T) \quad (25)$$

$$\theta_\rho^* = \min_{\theta_\rho} L_\rho(t, z, \hat{T}_p, \hat{\rho}_m; \theta_\rho) \quad (26)$$

Training the CPINN was completed in a sequential manner, training one of the networks at a time for a given number of epochs, whilst fixing the parameters of the other, and vice-versa. The sequential training approach was adopted as previous studies have shown that training separate networks simultaneously fails to reduce the loss function to a sufficient value (Amini Niaki et al., 2021). The process of training the CPINN can be summarised the following steps:

1. Fix the parameters for either $\mathcal{N}_T$ or $\mathcal{N}_\rho$.
2. Complete a sequential training step for $\mathcal{N}_T$ or $\mathcal{N}_\rho$.
 - (a) Conduct a forward pass through each network (Eqs. (11) and (12)) to evaluate $\hat{T}_p$ and $\hat{\rho}_m$.
 - (b) Calculate the appropriate thermal and physical properties using $\hat{T}_p$ and $\hat{\rho}_m$ and the correlations specified in Table 1 to compute L_T or L_ρ .
 - (c) Evaluate the necessary gradients required to compute L_T or L_ρ using automatic differentiation (Baydin et al., 2018).
 - (d) Calculate L_T or L_ρ using Eq. (19) or Eq. (24).

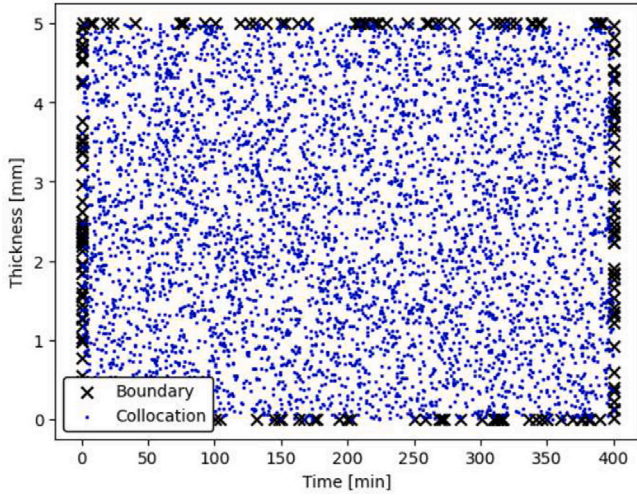


Fig. 3. Generation of training points for the domain, defined as collocation points, and points for the boundary and initial conditions, defined as boundary points using latin hypercube sampling methods. To improve readability every 10th collocation point and every 40th boundary point is displayed.

- (e) Train $\mathcal{N}_T$ or $\mathcal{N}_\rho$ using back-propagation to determine the partial gradients of the L_T or L_ρ with respect θ_T or θ_ρ and update the weights using the Adam Optimiser (Kingma and Ba, 2017).
 - (f) Repeat 10 times.
3. Stop after 600 sequential repeats of both $\mathcal{N}_T$ and $\mathcal{N}_\rho$. Note the number of repeats was chosen in line with reducing CPINN solution error in relation to the numerical solution.

2.5. Dataset generation

Training data was randomly sampled from the domain, known as collocation points within the bounds of the model, $t \in [0, 400 \text{ min}]$ and $z \in [0, 5 \text{ mm}]$. Concatenations of collocation points, $[t, z]$, were generated using a two dimensional latin hypercube sampling method available from the SciPy Quasi-Monte Carlo package. This generates a $n_c \times d$ array of values in $[0, 1]$ which were then transformed to the ranges required by multiplication by $[t_f, L]$, respectively. The number of collocation points, $n_c = 40,960$, was used to provide adequate data points for training the CPINN within the domain.

Similarly to the collocation points, collections of boundary points were generated using a two dimensional latin hypercube for the inlet, outlet and initial conditions. Here, $n_b = 4,096$ and $d = 2$, and for the inlet, outlet and initial conditions values were transformed from values in $[0, 1]$ by multiplication by $[t_f, 0 \text{ mm}]$, $[t_f, L]$ and $[0 \text{ s}, L]$, respectively. A summary of the data sampling is shown in Fig. 3.

2.6. Model construction

The coupled PINN architecture was constructed using TensorFlow 2.10 (Abadi et al., 2016) Keras deep learning API. The models relied on training and inference using an NVIDIA RTX A5000 GPU, which was facilitated using the CUDA[®] Toolkit 11.8. The network structure for each of the networks, $\mathcal{N}_T(t, z; \theta_T)$ and $\mathcal{N}_\rho(t, z; \theta_\rho)$, was identical and the structure of each individual network was based on optimised networks for similar problems (Zobeiry and Humfeld, 2021). To describe the structure, let $\{I, H \times D, O\}$ represent the network, where I is the input width, H is the hidden width, D is the hidden depth and O is the output width (Ian Goodfellow et al., 2016). For the drying model, the structure of both networks was $\{2, 32 \times 7, 1\}$ and the number of parameters for each CPINN was 6465.

3. Results and discussion

3.1. Model training

In this first instance, without transfer learning, the models were trained using a sequential methodology (Amini Niaki et al., 2021), training the heat balance network, $\mathcal{N}_T(t, z; \theta_T)$, for 10 epochs whilst keeping the parameters of the moisture balance network, $\mathcal{N}_\rho(t, z; \theta_\rho)$, fixed and vice-versa. The process was repeated 600 times, which was deemed sufficient to minimise the solution error, as shown by Figs. 4(a) and 4(b), whilst limiting training time. A batch size of 4,096 collocation points was used to compute the domain loss functions and all the boundary points were used to compute the boundary and initial condition loss functions at each batch. The Adam optimisation algorithm, a variant of stochastic gradient descent, was used to improve the trainable parameters with respect to the loss functions. The learning rate was initially chosen as 5×10^{-4} , whilst the parameters to assess the first and second moment estimate were the default TensorFlow values of 0.9 and 0.999 (Kingma and Ba, 2017). After 400 epochs, the learning rate was reduced to 1×10^{-4} to prevent stagnation of the learning process. The learning rates were chosen in line with those used for similar problems (Zobeiry and Humfeld, 2021; Amini Niaki et al., 2021) and verified by the fact the solution was similar to the numeric solution, as shown in Figs. 4(a) and 4(b). To initialise the parameters, without transfer learning, of the network the Xavier uniform initialiser was used for the weight whilst the biases were set to zero. The minimisation of the loss functions for each network is shown by the curves in Figs. 4(a) and 4(b).

The regularisation terms for the network were chosen for the heat balance as, $\lambda_{T,z=0} = 1 \times 10^{-6}$, $\lambda_{T,z=L} = 1 \times 10^{-6}$ and $\lambda_{T,t=0} = 1 \times 10^{-4}$ and for the moisture balance as, $\lambda_{\rho,z=0} = 1 \times 10^6$, $\lambda_{\rho,z=L} = 1 \times 10^6$, $\lambda_{\rho,t=0} = 1 \times 10^{-4}$. The values provided a similar order of magnitude for each loss function, as shown in Figs. 4(a) and 4(b), ensuring that the simulation physics were adhered to at the collocation and boundary points.

The CPINN networks were highly coupled during the sequential training procedure irrespective of the separate loss functions due to the fundamental physical equations. Taking the heat balance network training as an example, the heat loss functions are dependent on predicted physical and thermal properties, for example $\hat{k}_p$, which are in turn dependent on the predictions of both $\hat{T}_p$ and $\hat{\rho}_m$ as shown by the formulation of the correlations displayed in Table 1. Similarly to numeric methods, certain terms cause the heat and moisture balances to be highly coupled, for example, $\hat{S}_m$, had a large impact on the solution in general, appearing in both the heat and moisture outlet boundary conditions. As the performance of the network solution was dependent on the minimisation of the loss function, specifically, the outlet loss functions, the trainable parameters of the networks, θ_T and θ_ρ , both adapted to ensure the outlet condition was adhered to through correct prediction of $\hat{S}_m$.

3.2. Solution validity

The solution for $\hat{T}_p$ and $\hat{\rho}_m$, where $\hat{\rho}_m$ is represented by $\hat{X}_m$, using the CPINN is shown in Fig. 5(a) and (b), respectively. These figures highlight the complexity, non-linear and coupled nature of the solution and that the CPINN can effectively produce a complex mapping from t and z to $\hat{T}_p$ and $\hat{\rho}_m$ using the training methods and CPINN architecture outlined. To produce this solution and the following error comparisons, t and z validation points were sampled from those used for the numeric solution, effectively acting as a testing set. Due to the significant number of time points used in the numeric solution, 1000 time points were sampled at regular intervals and all z points were used, creating a solution map at the 200,000 points.

Figs. 5(a) and 5(b) shows that the CPINN correctly simulates the initial decrease in $\hat{T}_p$ and $\hat{X}_m$ at $t \sim 0 \text{ min}$ due to moisture loss and

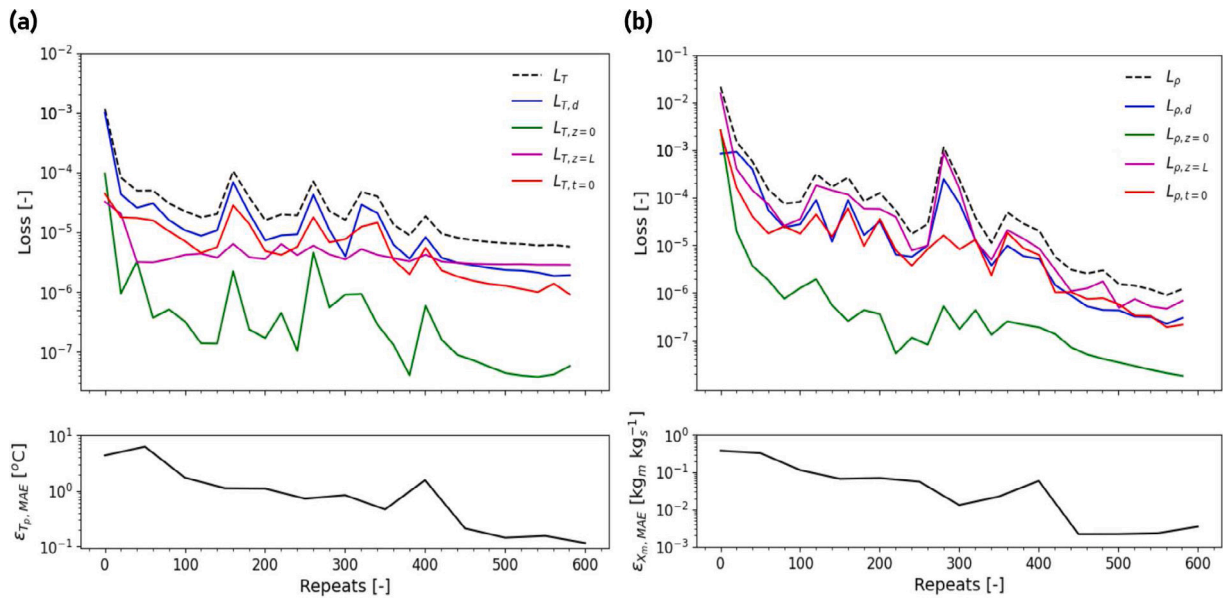


Fig. 4. Main figure: (a) Change of the heat loss functions and (b) moisture loss functions during sequential training of the CPINN, showing the change in domain, boundary condition and initial condition loss terms. Sub figure: Absolute mean error between the network and numeric solution for (a) $\hat{T}_p$ and (b) $\hat{X}_m$ during training at the sampled validation points.

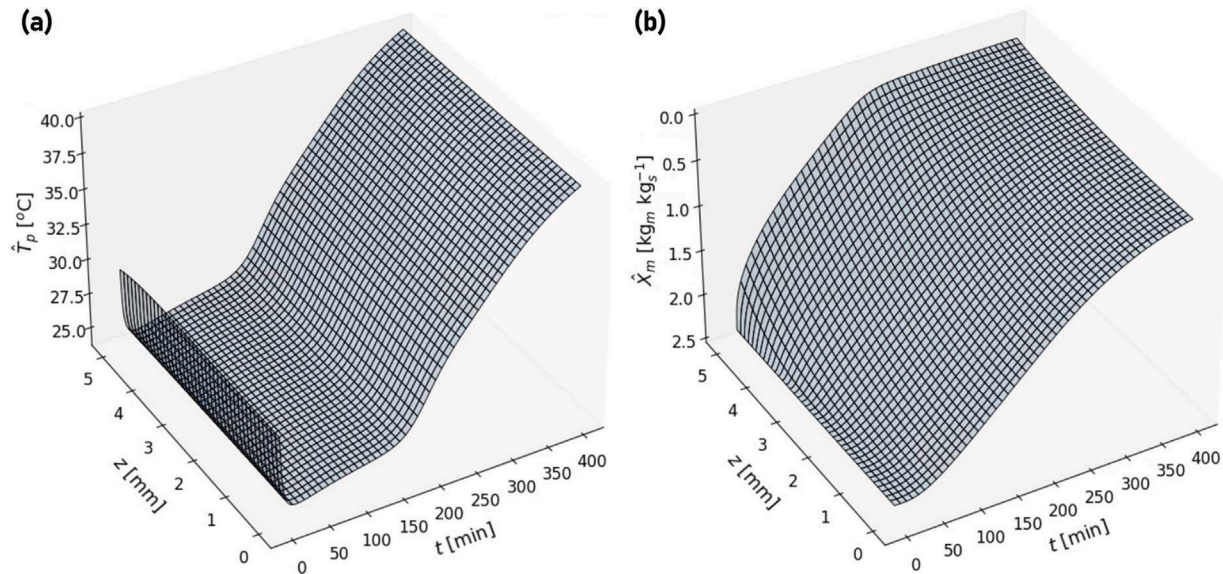


Fig. 5. Solution of the liquid diffusion controlled drying simulation using the CPINNs for (a) $\hat{T}_p$ and (b) $\hat{X}_m$ as a function of t and z for key drying conditions of $X_{m,t=0} = 2.35$, $T_{p,t=0} = 30^{\circ}\text{C}$ and $T_a = 45^{\circ}\text{C}$.

evaporative heat transfer at the drying surface ($z = L$). Additionally, after the initial and steep decline in $\hat{T}_p$ and $\hat{X}_m$ near the drying surface, the CPINN predicts the slow increase in $\hat{T}_p$ as the rate of loss of moisture reduces, representing a balance between convective heat transfer from the air and evaporative losses. Finally, at $t \sim 200$ min, the rate of $\hat{T}_p$ increase rises, as evaporative heat losses are reduced due to a reduction in moisture within the drying product. At $t = 400$ min, the product is almost dried, with the value of $\hat{X}_m$ approaching $0 \text{ kg}_m \text{ kg}_s^{-1}$ at all depths. At all time points, $\hat{X}_m$ decreases from $z = 0 \text{ mm}$ to $z = L$, as moisture diffuses towards the drying surface, replacing losses to the drying air. These phenomena are successfully described by the CPINN, representing the ability of the CPINNs to predict the complex dependencies between a system of coupled equations and changing thermo-physical properties.

The validity of the solution across the sampled t and z validation points was computed by finding the absolute error between $\hat{T}_p$ and $\hat{X}_m$ from the CPINN solution and the T_p and moisture content, X_m , from the numeric solution and plotted as an error contour plot as shown in Figs. 6(a) and 6(b). Overall, the solution of the CPINN is in agreement with the numeric solution, with errors in a few regions above 0.40°C difference for $\hat{T}_p$ and above $0.016 \text{ kg}_m \text{ kg}_s^{-1}$ for $\hat{X}_m$. Additionally, the mean absolute error, ϵ_{mae} , of the validation points for $\hat{T}_{p,z=L}$ and $\hat{X}_{m,z=L}$ was found to be $\epsilon_{T,mae} = 0.12^{\circ}\text{C}$ and $\epsilon_{\rho,mae} = 0.0035 \text{ kg}_m \text{ kg}_s^{-1}$, showing a globally low deviation from the numeric method.

The absolute error for $\hat{T}_p$ was greatest between $225 < t < 325$ min which correlated to the region where the rate of $\hat{T}_p$ change with respect to time increased and at the initial condition, due to a steep decrease in $\hat{T}_p$. The absolute error for $\hat{X}_m$ was greatest at the outlet boundary, specifically at regions of changing phenomena, for example

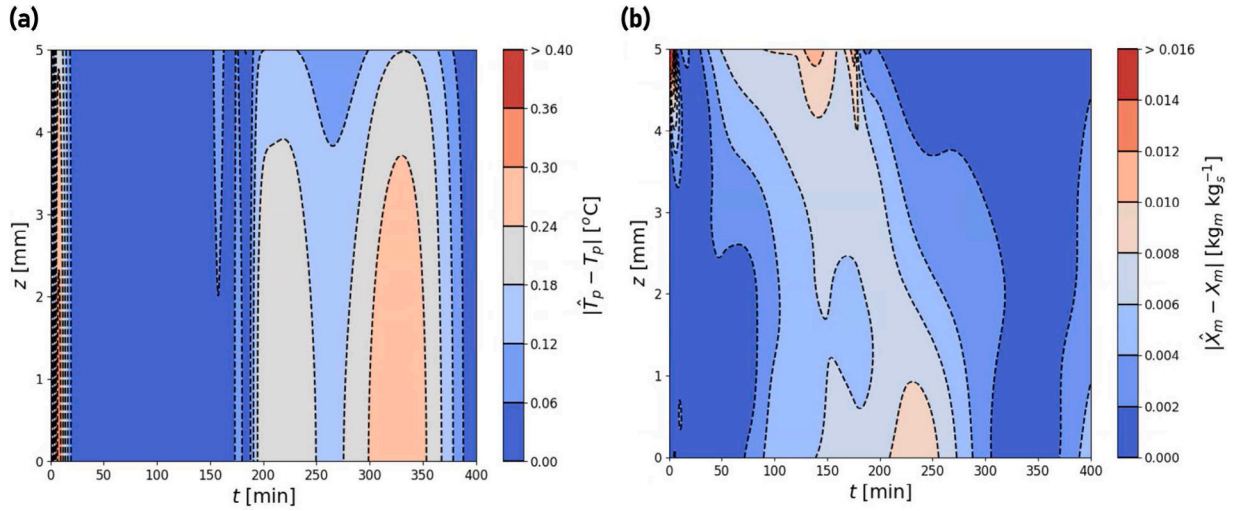


Fig. 6. Absolute error as a function of t and z , showing the difference between the (a) T_p numeric and network solutions and the (b) X_m numeric and network solutions for the entire solution domain for key drying conditions of $X_{m,t=0} = 2.35$, $T_{p,t=0} = 30$ °C and $T_a = 45$ °C.

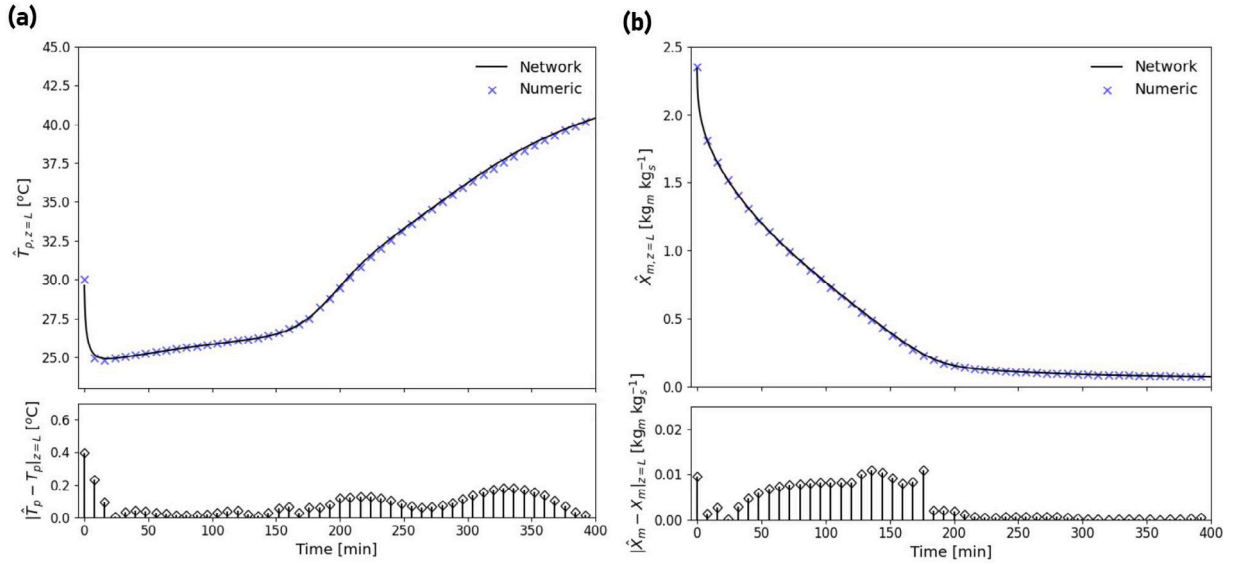


Fig. 7. Main figure: Comparison of the numerical (blue crosses) and CPINN (black line) solutions at every 20th validation point for (a) $\hat{T}_{p,z=L}$ and (b) $\hat{X}_{m,z=L}$ for key drying conditions of $X_{m,t=0} = 2.35$, $T_{p,t=0} = 30$ °C and $T_a = 45$ °C. Sub figure: Absolute error at every 20th time point comparing the numeric solution and CPINN solution (a) product temperatures and (b) product moisture content shown as stem points.

at $t \sim 0$ min and $t \sim 125$ – 200 min, representing initial and rapid losses in moisture and reductions in rate of moisture loss, respectively. In summary, higher error was correlated with regions of rapid solution variable change.

Comparisons of the numeric and CPINN solutions for $\hat{T}_{p,z=L}$ and $\hat{X}_{m,z=L}$ are shown in Fig. 7, respectively. The validity of the solution at the drying surface was of interest as this was where the solutions of $\hat{T}_{p,z=L}$ and $\hat{X}_{m,z=L}$ are strongly coupled through $\hat{S}_m$. The error of $\hat{T}_{p,z=L}$ was below 0.4 °C at all validation points at the drying surface, with greater errors between $225 < t < 325$ min for the aforementioned reasons. The error of $\hat{X}_{m,z=L}$ was below 0.012 $\text{kg}_m \text{kg}_s^{-1}$ at all validation points at the drying surface and the greatest errors occurred during initial moisture loss at $t \sim 0$ – 100 min.

3.3. Solver performance and transfer learning

To work as viable solvers the CPINN must outperform traditional explicit numeric methods in terms of solver time whilst maintaining

solution accuracy. In this section the solution times of the CPINN, t_{cpinn} , and fourth order Runge Kutta, t_{rk} , numeric methods are compared in addition to the global solution error as defined by mean absolute errors. The solver times for the CPINN included the training and inference time which was compared to the numeric solution through a time reduction, t_r , fraction as described in Eq. (27). If t_r was below one it represented an improvement over the numeric method.

$$t_r = \frac{t_{cpinn}}{t_{rk}} \quad (27)$$

Training the CPINNs using the sequential methodology and solving for the solution at the validation points in the first instance, now called Model #1, required the greatest time, $t_{cpinn} = 321$ min, as θ_T and θ_ρ were initialised using the generalised Xavier uniform methodology. The numeric solution time was approximately constant, $t_{rk} = 82.1$ min, for all solutions as at each new condition the full system had to be re-solved due to the explicit nature of the solver; in this first instance, $t_r = 3.9$.

To improve upon the solution times of the CPINN a staged transfer learning approach was adopted, initialising the parameters of a new

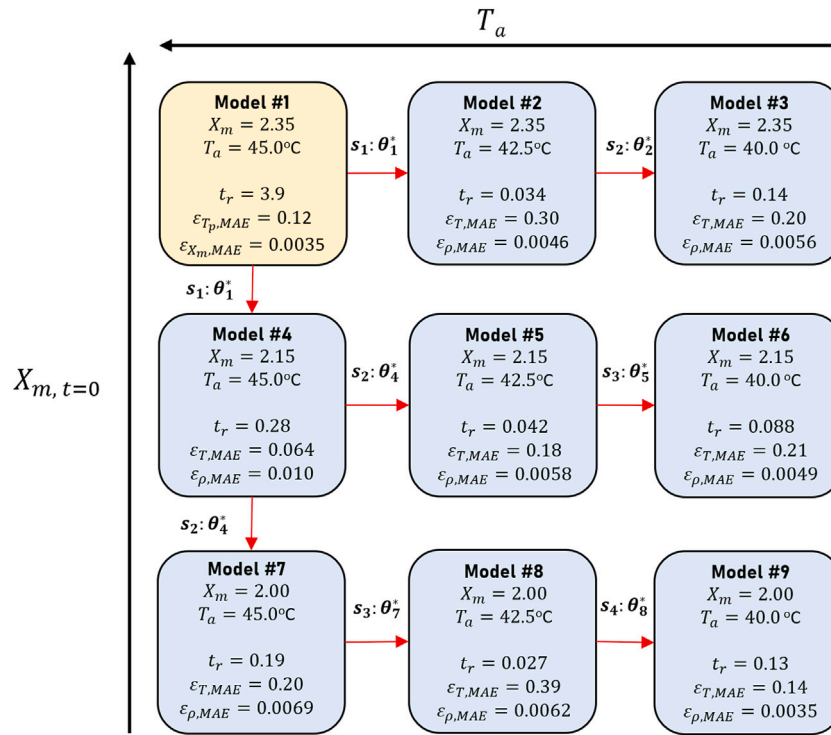


Fig. 8. Depiction of the transfer learning strategy to create a library of CPINNs. The first trained CPINN model, Model #1, is shown in yellow and subsequent models trained using a stage transfer learning approach are shown in blue. Each square summarises a models validity and performance through ϵ_{mae} and t_r at different key drying conditions, $X_{m,t=0}$ and T_a . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

CPINN with the parameters of a pre-trained CPINN with similar simulation conditions. The full staged transfer learning process is shown in Fig. 8 for nine models. The new CPINN was trained on a new set of key drying conditions, in this instance $X_{m,t=0}$ and T_a were varied, to mimic an optimisation process that would be undertaken on key drying conditions. For example, at stage 1, s_1 , the parameters of the Model #1 were used to initialise Model #2 and Model #4. Model #2 was trained based on a reduction in T_a by 2.5 °C and Model #4 was trained based on a reduction of $X_{m,t=0}$ by 0.20 kg_m kg_s⁻¹. Therefore, a transfer stage represents the use of optimal parameters from a CPINN model #i, defined as θ_i^* , trained at similar key drying conditions to initialise a new models parameters, θ_{i+1} . The new model is then trained on a new simulation with a change only in a single key drying conditions in order to improve solution time, where transfer stages were chosen with a smaller deviation in key drying conditions in order to reduce t_{cpinn} .

To train the CPINNs initialised with pre-trained CPINN parameters the sequential training methodology was once again used, employing the Adam Optimiser, a learning rate of 1×10^{-4} , the same regularisation terms, until both L_T and L_ρ were below 1×10^{-5} , as these loss values corresponded with a good solution when compared to the numeric solution as shown by Figs. 4(a) and 4(b).

Fig. 8 summarises t_r , $\epsilon_{T,mae}$, $\epsilon_{\rho,mae}$ for each CPINN model. The staged transfer of parameters from model to model provided significant improvement in solution time when compared to the numeric solution. For example, the t_r ranged from 0.28–0.027, whilst maintaining a similar order of $\epsilon_{T,mae}$ and $\epsilon_{\rho,mae}$, meaning that the CPINN solution was between 3 to 37 times faster. The worst case $\epsilon_{T,mae}$ and $\epsilon_{\rho,mae}$ was 0.39 °C and 0.01 kg_m kg_s⁻¹, respectively, indicating that the solution was both valid and performant for the chosen L_T and L_ρ limits.

The staged approach could easily be applied to a large range and dimension of key drying conditions. At each stage the model was saved which created a library of CPINNs. Once a library of CPINNs was created, the solution did not need to be saved, only the CPINN model itself, allowing the solution to be solved at run time by choosing

the correct CPINN for the desired key drying conditions. This is in contrast to numeric methods, where a sample of t , z , T_p and X_m must be saved from a previously simulated set of key drying conditions. For example, saving a numeric solution in csv format based on a map of 1000 sampled t and 200 sampled z validation points, and therefore two matrices of 1000×200 for the solution variables T_p and X_m had a memory footprint of 7 MB whilst saving a CPINN model was 0.8 MB. This shows that CPINNs effectively act as a method to compress solutions of PDEs. Additionally, any new combination of t and z , within the original trained domain limits, can be used as model inputs to CPINNs, meaning the constraint of saving specific sampled points is removed. Explicit numeric methods take substantial file space to store for future evaluation typically because of the small time step. CPINNs can be stored for future evaluation in a fraction of the file space.

4. Conclusions

A CPINN was successfully used to solve a system of coupled equations describing liquid diffusion controlled drying of a food product under a low temperature drying air. The architecture of the CPINN was constructed to provide predictions of the solution variables, $\hat{T}_p$ and $\hat{X}_m$, thermo-physical properties and by automatic differentiation, relevant gradients and fluxes. Additionally, the CPINN captured the impacts of the moisture loss which cause the solution to be highly coupled through a moisture sink term and evaporative heat loss in the moisture and thermal balances, respectively. The ϵ_{MAE} for the benchmark simulation, defined at a T_a of 45°C and an $X_{m,t=0}$ of 2.35 kg_m kg_s⁻¹, was 0.12 °C and 0.0035 kg_m kg_s⁻¹, respectively. At these conditions, the CPINN showed the product drying and at $t = 400$ min, the product was almost dried, with a value close to $\hat{X}_m = 0$ kg_m kg_s⁻¹ at all depths.

Initial training of the CPINN using the sequential training method was the rate limiting step, required the greatest solution time when compared to the numeric solution with $t_r = 3.9$, due to generalised initialisation. However, by utilising a staged transfer learning approach, solving the drying problem at different key drying conditions provided

a t_r , from 0.28–0.027 representing a 3 to 37 times faster solution when compared to numeric solutions, whilst maintaining a reasonable $\varepsilon_{T,mae}$ and $\varepsilon_{\rho,mae}$ with worst case values of 0.39 °C and 0.01 kg_m kg⁻¹. By saving a library of CPINNs various key drying conditions can be quickly simulated in future, as the CPINN can be evaluated at runtime allowing the CPINN to effectively act as a method to compress solutions of PDEs using a meshless approach.

The novelty of the CPINN architecture and library transfer learning approach served as an accelerated modelling approach of generalised transient and coupled PDEs additionally providing correct predictions of thermo-physical properties and important derivatives at run time. This work provides a benchmark for further studies, adding complexity in terms of multiscale, multiphase and multidimensional studies. Additionally, the architecture of the CPINN permits a physics informed approach and does not solely rely on training the model based on previously obtained numeric or experimental validation data, which maybe incorrect and invalidate models trained without a physics based approach.

In future, the training time of the CPINNs could be improved by optimising the number of sample points, learning rates and width and depth of hidden layers, additionally the regularisation terms for the loss functions could be determined by a structured method, for example, the method proposed by Wang S. et al. 2020 automatically adapts the regularisation terms based on individual loss gradients (Wang et al., 2020). Additionally the CPINN could be extended to include the impacts of shrinkage by applying an appropriate model within the loss function. Finally, the transfer learning methodology could be further investigated, for example models could be trained in parallel to further improved average training times or by optimising which parameter sets used to initialise in CPINN could be studied.

Nomenclature

Roman	
A_w	Water activity, –
$\mathbf{b}$	Bias vector, –
C_p	Product specific heat capacity, kJ kg ⁻¹ K ⁻¹
D	Hidden layer depth, –
D_w	Moisture diffusivity, m ² s ⁻¹
d	Dimensionality, –
D	Hidden layer width, –
h_a	Air heat transfer coefficient, W m ⁻² K ⁻¹
h_{fg}	Latent heat of vaporisation, kJ kg ⁻¹
h_m	Mass transfer coefficient at the drying surface, m s ⁻¹
h_v	Total enthalpy of vaporisation, kJ kg ⁻¹
Δh_w	Heat of wetting, kJ kg ⁻¹
I	Input layer width, –
k_a	Air thermal conductivity, W m ⁻¹ K ⁻¹
k_p	Product thermal conductivity, W m ⁻¹ K ⁻¹
L	Drying product thickness, m
L_i	Loss function, <i>var</i>
$\mathcal{N}_T$	Product temperature neural network, –
$\mathcal{N}_\rho$	Moisture density neural network, –
O	Output layer width, –
M_w	Moisture molecular weight, kg mol ⁻¹
N	Number of samples, –
n_d	Number of boundary sample points, –
n_c	Number of collocation sample points, –
Nu	Nusselt number, –
Pr	Prandtl number, –
P_{sat}	Saturation vapour pressure, Pa
q_c	Convective heat transfer, W m ⁻²
q_e	Evaporative heat transfer, W m ⁻²

R	Universal gas constant, Pa m ³ mol ⁻¹ K ⁻¹
Re	Reynolds number, –
Sc	Schmidt number, –
Sh	Sherwood number, –
S_m	Moisture sink, kg m ⁻² s ⁻¹
t	Time, s
T_a	Air temperature, K
t_f	Final time, s
T_p	Product temperature, K
t_r	Time reduction fraction, –
u_a	Air velocity, m s ⁻¹
X_m	Moisture content, kg _m kg _s ⁻¹
z	Drying product depth, m
$\mathbf{W}$	Weight matrix

Greek	
ε_{MAE}	Mean absolute error, <i>var</i>
θ_T	Product temperature network parameters, –
θ_ρ	Moisture density network parameters, –
λ	Regularisation term, –
μ_a	Air viscosity, Pa s
ρ_m	Moisture density, kg m ⁻³
ρ_p	Product density, kg m ⁻³
ρ_s	Dry product density, kg m ⁻³
σ	Activation function, –
ψ	Relative humidity, [%]
Abbreviations	
CPINN	Coupled physics informed neural network
ODE	Ordinary differential equation
PDE	Partial differential equation
PINN	Physics informed neural network
Subscripts and Superscripts	
a	Air
m	Moisture
p	Product
s	Dry
$t=0$	Initial condition
$z=0$	Base plate boundary
$z=L$	Drying surface boundary
$\wedge$	Network prediction
$*$	Optimal

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.fbp.2024.02.004>.

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